expansion of the SDE method. The series expansion is therefore not particularly informative on its own, but only in relation to its action on some particular distribution.

A simple first order example of a Langevin dynamics integrator is the method obtained by composing one of the Symplectic Euler variants with an Ornstein-Uhlenbeck step. To get a feel for how the expansion goes, let us work out its terms for such a splitting scheme in the case of a one degree of freedom model with unit mass: $H = p^2/2 + U(q)$. To be explicit, let us say our numerical method first solves exactly the OU process, then evaluates the impulse due to the force $-U'(q)$ and lastly drifts in position space according to the momentum. This method is denoted in our notation as $[\![\mathrm{OBA}]\!]$, where the evolution of measure in one step can be computed by writing

$$\exp(h\hat{\mathcal{L}}^\dagger_{[\![\mathrm{OBA}]\!]}) = \exp(h\mathcal{L}^\dagger_{\mathrm{A}})\exp(h\mathcal{L}^\dagger_{\mathrm{B}})\exp(h\mathcal{L}^\dagger_{\mathrm{O}}).$$

In this one dimensional example, we substitute

$$Z = \mathcal{L}^\dagger_{\mathrm{A}} = -p\frac{\partial}{\partial q}, \qquad Y = \mathcal{L}^\dagger_{\mathrm{B}} = U'(q)\frac{\partial}{\partial p},$$
$$X = \mathcal{L}^\dagger_{\mathrm{O}} = \gamma\left(\mathrm{Id} + p\frac{\partial}{\partial p}\right) + \frac{\sigma^2}{2}\frac{\partial^2}{\partial p^2},$$

into (7.18), where $\sigma = \sqrt{2\gamma k_{\mathrm{B}}\mathrm{T}}$. This gives us an expansion for the Kolmogorov operator for our discretized map

$$\hat{\mathcal{L}}^\dagger_{[\![\mathrm{OBA}]\!]} = S = \mathcal{L}^\dagger_{\mathrm{A}} + \mathcal{L}^\dagger_{\mathrm{B}} + \mathcal{L}^\dagger_{\mathrm{O}} + \frac{h}{2}\left([\mathcal{L}^\dagger_{\mathrm{A}}, \mathcal{L}^\dagger_{\mathrm{B}}] + [\mathcal{L}^\dagger_{\mathrm{A}}, \mathcal{L}^\dagger_{\mathrm{O}}] + [\mathcal{L}^\dagger_{\mathrm{B}}, \mathcal{L}^\dagger_{\mathrm{O}}]\right) + \ldots$$

It is clear that even writing out the $\mathcal{O}(h^2)$ terms considerably increases the complexity of the right hand side.

The commutators are best considered in relation to their action on a function (or rather, their action on a density) $\rho(q,p)$. For example, we may compute directly

$$\begin{aligned}[\mathcal{L}^\dagger_{\mathrm{A}}, \mathcal{L}^\dagger_{\mathrm{B}}]\rho &= -p\frac{\partial}{\partial q}\left(U'(q)\frac{\partial \rho}{\partial p}\right) - U'(q)\frac{\partial}{\partial p}\left(-p\frac{\partial \rho}{\partial q}\right)\\ &= -p(U'(q)\rho_p)_q + U'(q)(p\rho_q)_p,\\ &= -pU''(q)\rho_p - pU'(q)\rho_{pq} + U'(q)\rho_q + U'(q)p\rho_{qp}.\end{aligned}$$

As this is the commutator of two 1st order differential operators, the mixed partial term vanishes, leaving

$$[\mathcal{L}^\dagger_{\mathrm{A}}, \mathcal{L}^\dagger_{\mathrm{B}}]\rho = -pU''(q)\rho_p + U'(q)\rho_q.$$